

BOLNISI, GeorgiaWMO: **376210**Lat: **41.4509N**Lon: **44.5585E**Elev: **537**StdP: **95.04**Time Zone: **4.00 (E04)**Period: **05-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.5	-4.6	-13.2	1.3	-2.3	-11.1	1.5	-1.8	5.9	8.5	4.0	3.4	0.3	270	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.4	34.3	23.4	33.0	22.6	31.8	22.0	25.2	32.0	24.1	31.4	23.0	30.3	1.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.3	19.3	30.1	21.7	17.5	28.9	20.6	16.3	27.9	80.8	31.8	75.6	31.3	71.3	30.4	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.1	2.7	2.0	-7.9	36.3	3.1	1.3	-10.1	37.2	-11.9	38.0	-13.6	38.7	-15.9	39.7		
(5)			-9.1	25.9	2.7	1.8	-11.1	27.2	-12.7	28.3	-14.2	29.3	-16.2	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	2.4	3.7	8.0	12.4	17.7	22.7	25.4	25.4	20.4	14.4	7.8	4.0
	DBStd	8.84	3.52	3.87	3.33	3.74	3.02	2.80	2.69	2.94	3.36	3.31	3.55	3.14
	HDD10.0	787	237	178	79	18	1	0	0	0	0	6	82	187
	HDD18.3	2356	494	411	319	178	47	2	1	1	17	126	317	444
	CDD10.0	2159	1	2	18	91	239	382	476	478	313	143	15	1
	CDD18.3	686	0	0	0	2	28	133	218	220	79	4	0	0
	CDH23.3	6586	0	0	1	19	183	1139	2251	2383	589	21	0	0
	CDH26.7	2685	0	0	0	1	23	386	997	1111	166	1	0	0
Wind	WSAvg	0.7	0.6	0.7	0.9	0.7	0.7	0.9	0.9	0.9	0.7	0.6	0.6	0.6
Precipitation	PrecAvg	577	23	28	43	76	84	82	44	37	46	55	39	22
	PrecMax	734	54	63	100	147	143	154	89	109	125	138	94	80
	PrecMin	452	3	6	3	24	27	27	14	6	9	14	3	1
	PrecStd	82	14	16	26	32	29	34	20	22	29	29	25	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	17.1	20.2	25.6	29.1	33.6	36.2	36.2	32.3	25.5	18.7	15.1
		MCWB	6.9	8.8	13.8	18.2	19.5	21.8	24.3	24.7	22.2	17.6	11.9	8.2
	2%	DB	11.2	13.9	17.8	23.2	26.7	31.9	34.2	34.6	30.4	23.3	16.6	12.5
		MCWB	5.5	7.6	11.5	15.8	18.3	21.6	23.3	23.9	21.4	16.0	10.6	6.8
	5%	DB	9.3	11.6	15.9	21.3	25.4	30.4	32.9	33.4	28.6	21.5	14.7	10.6
		MCWB	4.3	6.1	10.0	14.8	18.0	21.2	23.0	22.8	20.1	14.9	9.5	5.9
	10%	DB	7.4	9.5	14.1	19.3	23.8	29.0	31.6	32.2	26.9	19.9	13.0	8.9
		MCWB	3.2	4.9	8.9	13.5	17.1	20.6	22.2	22.0	19.1	14.2	8.7	5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	9.2	15.6	19.3	21.9	25.1	27.8	26.6	23.7	18.5	13.0	9.4
		MCDB	12.9	15.8	19.6	24.4	27.5	29.6	31.7	33.8	30.0	23.3	16.4	13.2
	2%	WB	5.9	7.8	12.2	16.9	19.7	23.3	25.5	25.2	22.1	17.0	11.6	7.7
		MCDB	9.7	13.1	16.1	21.6	24.7	29.4	31.5	32.6	29.1	21.6	15.0	11.2
	5%	WB	4.8	6.4	10.5	15.2	18.5	22.1	24.1	24.1	20.9	15.9	10.5	6.6
		MCDB	8.3	11.0	14.9	20.6	24.0	29.0	31.4	31.9	27.3	20.1	13.3	9.7
	10%	WB	3.7	5.2	9.2	13.9	17.4	21.0	23.1	22.9	19.7	14.8	9.4	5.5
		MCDB	6.8	8.8	13.4	18.6	23.0	27.9	30.4	30.6	25.6	18.9	12.3	8.4
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.5	8.7	9.0	9.3	10.1	10.2	10.4	9.3	7.8	7.4	7.3
		MCDBR	9.8	11.4	11.6	11.8	11.7	11.6	11.7	12.1	11.5	10.5	10.0	9.6
		MCWBR	6.4	7.2	7.5	7.0	6.4	5.6	5.8	6.3	6.0	5.6	6.1	6.2
	5% WB	MCDBR	8.6	10.4	10.4	11.4	10.6	11.1	11.0	11.6	10.7	9.3	8.3	8.0
		MCWBR	5.9	6.7	7.0	7.1	6.3	6.4	6.1	6.5	6.0	5.2	5.5	5.5
Clear-Sky Solar Irradiance	taub	0.327	0.366	0.417	0.485	0.453	0.441	0.484	0.479	0.423	0.402	0.349	0.324	
	taud	2.320	2.197	2.094	1.955	2.120	2.217	2.109	2.117	2.258	2.244	2.351	2.349	
	Ebn at Noon	809	824	821	789	828	838	796	787	809	775	777	780	
	Edh at Noon	88	115	144	178	154	140	155	149	121	109	85	79	
All-Sky Solar Radiation	RadAvg	1.91	2.86	3.50	4.34	5.24	6.18	6.03	5.53	4.33	2.88	2.03	1.61	
	RadStd	0.23	0.43	0.38	0.56	0.36	0.45	0.50	0.59	0.36	0.31	0.25	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page